

ORIGINAL ARTICLE

Immunological dysregulation and adverse perinatal outcomes in pregnant women with human papillomavirus infection

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ABSTRACT

Background and Aim: Human papillomavirus (HPV) infection during pregnancy can induce local immune alterations in the cervical mucosa, potentially affecting perinatal outcomes. However, data on humoral immune parameters in cervical mucus of pregnant women with transient and persistent HPV infection remain limited. The aim of this study was to evaluate immunoglobulin and cytokine profiles in cervical mucus and to assess their possible association with fetal and neonatal outcomes.

Methods: This prospective observational study included a subgroup of 77 pregnant women: 28 with transient HPV infection, 24 with persistent HPV infection, and 25 HPV-negative controls. Immunological evaluation of cervical mucus was performed, including assessment of immunoglobulins (IgG, IgA, IgM) and cytokine levels (IFN- α , IFN- γ , IL-1 β , IL-2, IL-4, IL-8, IL-10, TNF- α). Neonatal outcomes were also analyzed.

Results: Women with transient and persistent HPV infection showed significantly higher cervical IgG levels compared to controls ($p < 0.05$ and $p < 0.01$, respectively). Levels of IFN- α , IFN- γ , IL-1 β , IL-2, IL-4, and IL-8 were also significantly elevated in HPV-positive groups. These immunological changes were associated with worse neonatal outcomes, including lower Apgar scores in newborns from HPV-infected mothers.

Conclusions: Persistent and even transient HPV infection during pregnancy is accompanied by significant local humoral immune activation in cervical mucus. These alterations may contribute to adverse neonatal



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outcomes. Further studies are needed to determine whether intensified monitoring of HPV-positive pregnant women can improve perinatal results. (www.actabiomedica.it)

Key words: human papillomavirus, HPV infection, pregnancy, cervical mucus, local immunity, IgG, cytokines, immune dysregulation, neonatal outcomes

Introduction

Human papillomavirus (HPV) is one of the most common sexually transmitted viral infections worldwide and represents a significant public health concern for women of reproductive age (1,2). Although the oncogenic potential of high-risk HPV genotypes is well established (3), increasing attention has recently been paid to the impact of HPV on reproductive health and pregnancy outcomes (4,5).

Pregnancy is accompanied by substantial immunological adaptations, including modulation of both systemic and local mucosal immunity, aimed at maintaining fetal tolerance (5,6). These physiological changes can alter the natural history of HPV infection, often promoting viral persistence and affecting local immune responses at the cervical level (7,8).

Cervical mucus serves as a critical immunological barrier between the external environment and the upper genital tract. It contains a complex mixture of immunoglobulins and cytokines that actively participate in local defense mechanisms. Secretory immunoglobulin A (sIgA) is the predominant antibody on mucosal surfaces and provides the first line of defense by neutralizing pathogens and preventing their attachment to epithelial cells. In contrast, IgG reaches the cervical mucus primarily through transudation from serum and becomes particularly important during prolonged antigenic stimulation. In chronic viral infections, including persistent HPV, a shift toward elevated IgG levels is frequently observed, indicating activation of systemic humoral immunity and ongoing antigenic drive (7-9).

Despite the recognized importance of local mucosal immunity, studies evaluating immunoglobulin and cytokine profiles in cervical mucus during pregnancy in the context of transient versus persistent

HPV infection remain very limited. Understanding these local immune changes is essential, as they may represent one of the key mechanisms linking maternal HPV infection with adverse neonatal outcomes.

Several studies have demonstrated significant alterations in cytokine profiles among HPV-infected women; however, comprehensive data regarding humoral immune parameters (IgG, IgA, IgM) and cytokine levels in cervical mucus during pregnancy are still scarce, especially when comparing transient and persistent infection (8-10). Such local immune dysregulation may play an important role in the pathogenesis of adverse perinatal outcomes associated with HPV (4,5,11).

Previous studies conducted by our research group in Azerbaijan revealed a higher frequency of adverse pregnancy and neonatal outcomes in HPV-positive women (12). These findings provided the rationale for the present study, which focuses specifically on local immune alterations in cervical mucus.

The aim of this prospective observational study was to evaluate the profiles of immunoglobulins (including IgA and IgG) and cytokines in the cervical mucus of pregnant women with transient and persistent HPV infection and to assess their possible association with neonatal outcomes.

Patients and methods

Study design and setting

This prospective observational study was conducted between 2015 and 2022 at the Educational-Surgical Clinic and affiliated maternity centers of Azerbaijan Medical University, Baku, Azerbaijan. The

study was designed to evaluate immunological alterations, placental dysfunction, pregnancy complications, and adverse perinatal outcomes associated with human papillomavirus (HPV) infection during pregnancy.

Study population

The study was performed within a larger cohort of 206 pregnant women. For the immunological part of the study, a subgroup of 77 pregnant women was selected. This subgroup included 28 women with transient HPV infection, 24 women with persistent HPV infection, and 25 HPV-negative pregnant women (control group).

Inclusion and exclusion criteria

Women aged 17–40 years with singleton pregnancies were eligible for inclusion. HPV infection was confirmed by polymerase chain reaction (PCR) testing of cervical samples, with repeat testing performed after 6 months to differentiate transient from persistent infection. All participants gave written informed consent.

Exclusion criteria were multiple pregnancies, HIV infection, severe chronic somatic or endocrine diseases, acute systemic infections unrelated to HPV, and known fetal chromosomal abnormalities.

HPV diagnostics

HPV infection was diagnosed using PCR testing of cervical samples. Participants also underwent standard gynecological examination and colposcopy when clinically indicated.

Immunological evaluation

Detailed immunological assessment was performed on cervical mucus samples obtained from the 77 women. The analysis included measurement of immunoglobulin levels: secretory IgA (sIgA), IgA, IgM, and IgG. Cytokine profiling included interferon- α (IFN- α), interferon- γ (IFN- γ), interleukin-1 β (IL-1 β), interleukin-2 (IL-2), interleukin-4 (IL-4), interleukin-8 (IL-8), interleukin-10 (IL-10), receptor antagonist of interleukin-1 (RAIL-1), and tumor

necrosis factor- α (TNF- α). Additional parameters assessed were neutrophil functional activity, phagocytic activity, and lysosomal enzyme activity. HPV-specific IgG antibodies were also evaluated.

Neonatal outcomes

All newborns were evaluated immediately after delivery and during the early neonatal period. Neonatal assessment included Apgar scores at 1 and 5 minutes, birth weight, signs of intrauterine growth restriction, and presence of congenital malformations. Pregnancy and delivery data were collected as part of routine clinical follow-up.

Statistical analysis

Statistical analysis was performed using SPSS software (version 26.0, IBM Corp., Armonk, NY, USA). Quantitative variables are presented as mean $\pm$ standard deviation (SD). Differences between groups were compared using the Student's t-test or Mann-Whitney U test, as appropriate. Categorical variables were analyzed using the chi-square test or Fisher's exact test. A p-value of <0.05 was considered statistically significant.

Ethical considerations

The study protocol was approved by the Ethics Committee of Azerbaijan Medical University (Protocol No. 19, dated November 10, 2015). All procedures were conducted in accordance with the ethical principles of the Declaration of Helsinki. Written informed consent was obtained from all participants before enrollment in the study.

Results

Baseline clinical and reproductive characteristics

The immunological analysis was performed in a subgroup of 77 pregnant women: 28 with transient HPV infection, 24 with persistent HPV infection, and 25 HPV-negative controls. The mean maternal

age was comparable between the groups (27.2 ± 6.0 years in HPV-positive groups versus 28.3 ± 5.0 years in controls, $p > 0.05$). Women with HPV infection had a significantly higher frequency of unfavorable reproductive history and concomitant sexually transmitted infections compared to the control group (Table 1).

Immunological alterations in cervical mucus

Significant changes in local humoral immunity were observed in women with HPV infection. The concentration of IgG in cervical mucus was statistically significantly higher in the transient HPV infection group (0.42 ± 0.06 g/L, $p < 0.05$) and especially in the persistent HPV infection group (0.64 ± 0.03 g/L, $p < 0.01$) compared with the control group (0.23 ± 0.06 g/L) (Table 2).

Analysis of the cytokine profile of cervical mucus revealed a pronounced immune dysregulation in HPV-positive pregnant women (Table 3). Levels

of IFN- α , IFN- γ , IL-1 β , IL-2, IL-4, and IL-8 were significantly elevated in both transient and persistent HPV infection groups compared to controls. The most marked cytokine imbalance was observed in women with persistent HPV infection, indicating sustained antigenic stimulation and disruption of local immune homeostasis.

In addition, HPV-specific IgG antibody titers were significantly higher in women with persistent HPV infection compared to those with transient infection and controls ($p < 0.001$) (Figure 1).

Neonatal outcomes

Newborns from HPV-positive mothers demonstrated worse neonatal outcomes. The mean Apgar score at 1 minute was significantly lower in the HPV-positive group compared to controls (7.1 ± 0.4 vs 8.2 ± 0.1 points, $p < 0.05$). The frequency of congenital

Table 1. Baseline clinical and reproductive characteristics of the study population.

Variable	Transient HPV infection (n=88)	Persistent HPV infection (n=64)	Control group (n=54)	p-value
	27.2 ± 6.0	27.2 ± 6.0	28.3 ± 5.0	>0.05
History of medical abortions, n (%)	28 (31.8%)	21 (32.8%)	9 (16.7%)	<0.05
History of spontaneous miscarriage, n (%)	14 (15.9%)	11 (17.2%)	3 (5.6%)	<0.05
Cervical erosion, n (%)	33 (37.5%)	24 (37.5%)	7 (13.0%)	<0.01
Any sexually transmitted infection, n (%)	82 (93.2%)	61 (95.3%)	20 (37.0%)	<0.001

Note: Data are presented as mean $\pm$ SD or number (%).

Table 2. Humoral immune parameters of cervical mucus in examined pregnant women.

Parameters	Control group (n=25)	Transient HPV infection (n=28)	Persistent HPV infection (n=24)
sIgA, g/L	0.021 ± 0.006	0.025 ± 0.007	0.036 ± 0.005
IgA, g/L	0.10 ± 0.026	0.09 ± 0.017	0.14 ± 0.018
IgM, g/L	0.012 ± 0.002	0.027 ± 0.008	0.022 ± 0.003
IgG, g/L	0.23 ± 0.06	$0.42 \pm 0.06^*$	$0.64 \pm 0.03^{**}$

Note: Data are presented as mean $\pm$ standard deviation (M $\pm$ SD).

$p < 0.05$ compared with the control group, $^{**} p < 0.01$ compared with the control group

Table 3. Cytokine profile of cervical mucus in examined pregnant women.

Cytokines	Control group (n=25)	Transient HPV infection (n=28)	Persistent HPV infection (n=24)
IFN- α , pg/mL	784.41 $\pm$ 164.61	2384.93 $\pm$ 375.12**	1980.72 $\pm$ 403.73*
IFN- γ , pg/mL	802.50 $\pm$ 65.66	1163.27 $\pm$ 101.03*	1124.30 $\pm$ 102.12**
IL-1 β , pg/mL	4092.25 $\pm$ 727.84	6763.95 $\pm$ 1332.44**	5652.34 $\pm$ 1911.13*
RAIL-1, pg/mL	4326.84 $\pm$ 762.62	6185.69 $\pm$ 1046.62**	5723.65 $\pm$ 1163.16*
IL-2, pg/mL	178.67 $\pm$ 11.37	328.61 $\pm$ 41.38***	332.38 $\pm$ 79.36**
IL-4, pg/mL	282.51 $\pm$ 53.66	452.70 $\pm$ 76.96**	354.72 $\pm$ 68.26
IL-8, pg/mL	8903.22 $\pm$ 1718.08	14901.45 $\pm$ 224.31**	11194.25 $\pm$ 3583.04*
IL-10, pg/mL	212.44 $\pm$ 11.34	283.42 $\pm$ 37.10*	186.51 $\pm$ 14.32
TNF- α , pg/mL	381.32 $\pm$ 73.15	352.05 $\pm$ 33.67*	362.15 $\pm$ 69.45

Note: Data are presented as mean $\pm$ standard deviation (M $\pm$ SD) p<0.05 compared with the control group

** p<0.01 compared with the control group

*** p<0.001 compared with the control group

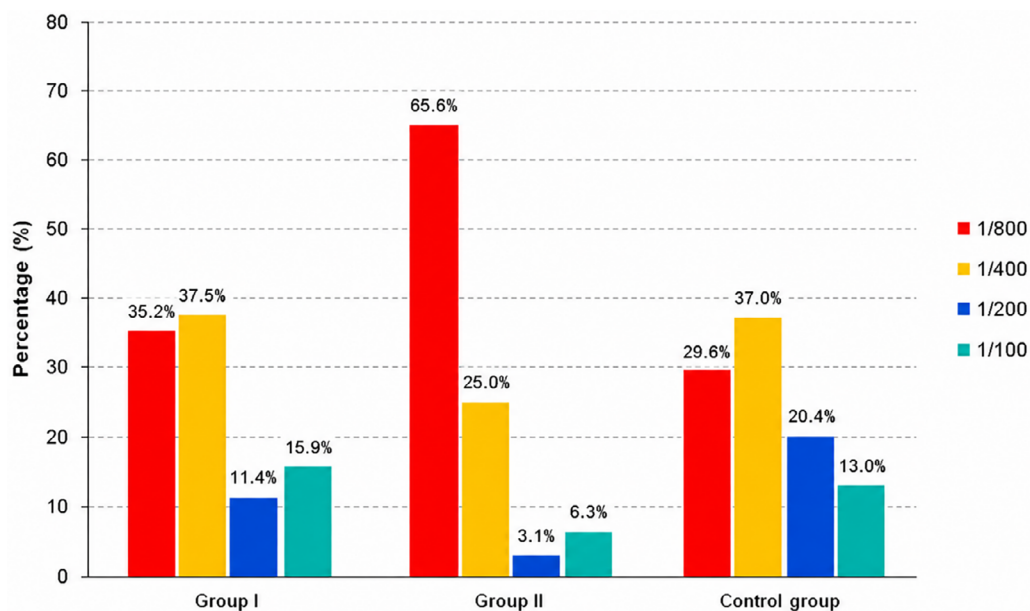


Figure 1. Comparative characteristics of HPV IgG antibody levels in pregnant women.

fetal malformations was also higher (25.0% vs 12.0%, p < 0.05), as was the rate of preterm birth (13.5% vs 4.0%, p < 0.05) (Table 4).

These findings suggest that local humoral immune dysregulation in the cervical mucus associated with HPV infection may contribute to adverse neonatal outcomes.

Discussion

The present study demonstrated that HPV infection during pregnancy is associated with significant alterations in the local cervical immune environment, particularly in women with persistent viral infection.

Table 4. Neonatal outcomes and congenital malformations.

Neonatal outcomes	HPV-positive women (n=52)	Control group (n=25)	p-value
Congenital malformations, n (%)	13 (25.0%)	3 (12.0%)	<0.05
Preterm birth, n (%)	7 (13.5%)	1 (4.0%)	<0.05
Apgar score at 1 min, points (M $\pm$ SD)	7.1 $\pm$ 0.4	8.2 $\pm$ 0.1	<0.05
Apgar score at 5 min, points (M $\pm$ SD)	8.0 $\pm$ 0.3	8.7 $\pm$ 0.1	NS

Note: Values are presented as mean $\pm$ standard deviation (SD) or number (%).

The most important findings included increased cervical IgG concentrations and pronounced changes in cytokine profiles, suggesting activation of both humoral and cellular immune responses. These observations support the concept that persistent HPV infection reflects inadequate viral clearance despite ongoing immune activation. Current evidence indicates that persistence, rather than transient HPV exposure, is the key determinant of adverse biological consequences and disease progression (1). Furthermore, HPV possesses multiple immune-evasion mechanisms that enable long-term survival within epithelial tissues despite activation of host defense pathways (3).

One of the principal findings of our study was the significant elevation of IgG levels in cervical mucus among HPV-positive pregnant women, particularly in those with persistent infection. Increased local IgG production likely reflects prolonged antigenic stimulation and activation of humoral immune defense mechanisms. However, the persistence of infection despite elevated immunoglobulin concentrations suggests that antibody-mediated responses alone are insufficient for complete viral elimination. Recent advances in HPV research emphasize the importance of complex interactions between viral persistence and host immune regulation in determining infection outcomes (2).

In addition to changes in humoral immunity, HPV-positive women demonstrated significantly elevated levels of IFN- α , IFN- γ , IL-1 β , IL-2, IL-4, IL-8, IL-10, and TNF- α in cervical mucus. The simultaneous increase in antiviral cytokines, including IFN- α and IFN- γ , indicates activation of host antiviral defense mechanisms. At the same time, elevated concentrations of IL-1 β , TNF- α , and IL-8 reflect persistent inflammatory stimulation within the cervical microenvironment.

Similar findings have been reported in recent studies describing HPV infection as a condition characterized by concurrent activation of inflammatory, antiviral, and immunoregulatory pathways (8).

The cytokine pattern observed in our study suggests a complex and dynamic immune response. Persistent HPV infection appears to induce chronic immune activation while simultaneously triggering regulatory mechanisms designed to limit excessive tissue damage. Elevated levels of IL-4 and IL-10 may represent compensatory anti-inflammatory responses aimed at maintaining tissue homeostasis. However, this balance may be insufficient to achieve effective viral clearance. Recent investigations of the cervical immune microenvironment have demonstrated that persistent HPV infection is associated with dysregulated cytokine signaling, altered immune-cell communication, and maintenance of a chronic inflammatory state that promotes viral persistence (7). These findings are consistent with our observation that women with persistent infection exhibited the most pronounced immunological alterations.

The importance of these immune disturbances may be particularly relevant during pregnancy. Successful pregnancy requires a carefully regulated balance between maternal immune defense and fetal tolerance. According to Condrat et al., HPV infection may disrupt this equilibrium and contribute to persistent immune activation at the maternal-fetal interface (8). Therefore, the cytokine imbalance identified in our study may have implications extending beyond the cervix and potentially influence placental development and function.

Indeed, HPV-positive women in our cohort demonstrated a higher frequency of placental abnormalities, including enlargement of the intervillous space,

vascular dilatation, increased placental thickness, and premature placental maturation. Although causality cannot be established, these findings are consistent with growing evidence suggesting that maternal HPV infection may adversely affect placental structure and function through inflammatory and immune-mediated mechanisms (5,9). Recent placental investigations have demonstrated the presence of HPV genetic material in placental tissues and reported associations between placental HPV infection and adverse pregnancy outcomes (11). Chronic inflammatory activation induced by persistent HPV infection may therefore contribute to impaired placental vascularization and altered fetoplacental homeostasis.

The placental abnormalities identified in our study may help explain the increased frequency of adverse obstetric outcomes observed among HPV-positive women. In particular, women with persistent HPV infection demonstrated higher rates of threatened miscarriage, premature rupture of membranes, preterm delivery, and neonatal complications. These findings are in agreement with previous systematic reviews and meta-analyses reporting significant associations between maternal HPV infection and adverse pregnancy outcomes, especially premature rupture of membranes and preterm birth (4,5,9,10). Although the exact biological mechanisms remain incompletely understood, chronic inflammation and placental dysfunction may represent important mediating factors.

An additional clinically relevant finding was the increased incidence of congenital abnormalities and impaired neonatal adaptation among infants born to HPV-positive mothers. Newborns from HPV-infected pregnancies more frequently demonstrated lower Apgar scores and signs of growth restriction. These outcomes may reflect the cumulative effects of chronic intrauterine inflammation, placental insufficiency, and altered maternal-fetal immune interactions. Similar associations between maternal HPV infection, placental pathology, and unfavorable neonatal outcomes have been reported in recent cohort studies (12).

Several limitations should be acknowledged. First, the observational design of the study precludes definitive conclusions regarding causality. Second, HPV genotyping and viral load assessment were not performed, preventing evaluation of the influence of specific viral

characteristics on immune responses and pregnancy outcomes. Third, immunological analyses were conducted in a limited subgroup of participants. These limitations have also been recognized in previous reviews addressing HPV infection during pregnancy (5,6,9).

In conclusion, our findings suggest that persistent HPV infection during pregnancy is associated with substantial alterations in cervical humoral immunity and cytokine regulation. The observed increase in IgG concentrations together with the imbalance of pro-inflammatory and anti-inflammatory cytokines indicates chronic local immune activation that may contribute to placental dysfunction and adverse neonatal outcomes. Further prospective studies incorporating HPV genotyping, viral load assessment, and comprehensive immunological profiling are needed to clarify the mechanisms linking persistent HPV infection with pregnancy complications and fetal development.

Limitations and future directions

The present study has several limitations. First, its observational design does not permit establishment of a direct causal relationship between HPV infection, local immune alterations, and adverse pregnancy outcomes. Second, immunological analyses were performed in a relatively small subgroup of participants, which may limit the generalizability of the findings. Third, HPV genotyping and quantitative viral load assessment were not conducted, preventing evaluation of the potential influence of specific HPV types and viral burden on immune responses and pregnancy outcomes. Finally, long-term follow-up of newborns was not available.

Despite these limitations, the study provides prospective evidence that persistent HPV infection during pregnancy is associated with substantial alterations in cervical humoral immunity and cytokine regulation. Future large-scale multicenter studies should incorporate HPV genotyping, viral load assessment, detailed placental molecular investigations, and comprehensive longitudinal follow-up of offspring. Such approaches may improve understanding of the mechanisms linking persistent HPV infection, immune dysregulation, placental dysfunction, and adverse perinatal outcomes.

Conclusions

Persistent human papillomavirus infection during pregnancy is associated with significant alterations in the local cervical immune environment. Women with persistent HPV infection demonstrated increased cervical IgG concentrations together with marked changes in cytokine expression, including elevated levels of IFN- α , IFN- γ , IL-1 β , IL-2, IL-4, IL-8, IL-10, and TNF- α , indicating activation of both inflammatory and antiviral immune pathways.

The observed immunological profile suggests that persistent HPV infection is characterized by chronic local immune activation accompanied by ineffective viral clearance. The simultaneous increase in pro-inflammatory and anti-inflammatory cytokines may reflect an imbalanced immune response that contributes to maintenance of viral persistence during pregnancy.

These immune alterations were accompanied by a higher frequency of placental abnormalities, including changes in placental morphology and vascular structure, suggesting a possible relationship between local immune dysregulation and placental dysfunction. Furthermore, HPV-positive pregnancies were associated with less favorable obstetric and neonatal outcomes, including increased rates of premature rupture of membranes, preterm delivery, lower neonatal adaptation scores, and congenital abnormalities.

Overall, the findings support the hypothesis that chronic immune dysregulation may represent an important mechanistic link between persistent HPV infection, placental pathology, and adverse perinatal outcomes. Early identification and careful monitoring of pregnant women with persistent HPV infection may facilitate risk stratification and improve maternal and neonatal care. Further studies are required to clarify the biological pathways underlying these associations and to identify potential targets for preventive interventions.

Ethical approval: The study was approved by the Ethics Committee of Azerbaijan Medical University (Protocol No. 19, November 10, 2015) and conducted in accordance with bioethical standards.

Conflict of interest: Each author declares that he or she has no commercial associations (e.g., consultancies, stock ownership, equity interest, patent/licensing arrangement, etc.) that might pose a conflict of interest in connection with the submitted article.

Authors' contribution: SI contributed to conceptualization, study design, patient recruitment, data collection, laboratory investigations, interpretation of results, and drafting of the manuscript. LM contributed to literature review, clinical data collection, and manuscript preparation. ES contributed to scientific supervision of the research process, methodological guidance, critical revision of the manuscript, and approval of the final version. SG contributed to methodology, statistical analysis, data visualization, manuscript editing, interpretation of data, and preparation of tables and figures. All authors have read and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Declaration on the use of AI: None.

Consent for publication: Written informed consent was obtained from all participating pregnant women and from the parents of the newborns prior to inclusion in the study. Participants were informed about the study procedures and the scientific use of anonymized data.

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